CHAPTER 1—Errors

A **number** is a fundamental mathematical concept used to represent **quantity**, **order**, or **measurement.** There are two kinds of numbers. Exact and Approximate numbers.

1.1 Exact Numbers

- ✓ These are values that are **precisely defined** and **do not change**.
- ✓ Examples include:
  - ➢ Whole numbers like 1, 2, 3
  - ➢ Rational fractions like $\frac{3}{4}, \frac{3}{5}$
  - ➢ Constants like $\pi, \sqrt{2}, e$, **when written symbolically** (not numerically approximated)

These are considered "true values" in mathematical computations.

1.1 Approximate Numbers

- ✓ These are **numerical representations** that are **close to** but **not exactly equal to** the true value.
- ✓ Used in practical computations where exact values are either unknown or too complex to use directly.
- ✓ Examples:
  - ➢ 1.41or 1.414as approximations of $\sqrt{2}$
  - ➢ $3.14, 3.141, 3.14159$as approximations of π

These approximations are necessary because many real-world problems involve irrational numbers or transcendental functions that cannot be expressed exactly in decimal form.

1.2 Significant Digits

The digits that are used to express a number are called significant digits.

Definition 1 A significant digit of an approximate number is any non-zero digit in its decimal representation or any zero lying between significant digits or used as place holder to indicate a retained place.

Definition 2 Significant digits are the digits in a number that carry meaningful information about its precision. They help us understand how accurate or reliable a number is, especially when it's an approximation

The digits 1, 2, 3, 4, 5, 6, 7, 8, 9 are significant digits. '0' is also a significant figure except when it is used to fix the decimal point, or to fill the places of unknown or discarded digits. For example, in the number 0.0005010, the first four '0's' are not significant digits, since they serve only to fix the position of the decimal point and indicate the place values of the other digits. The other two '0's' are significant.

A digit is **significant** if it is:

- ✓ A **non-zero digit** (1–9)
- ✓ A **zero** that:
  - ➢ Lies **between significant digits** (e.g., in 105, the zero is significant)
  - ➢ Is **to the right of the decimal point** and **to the right of a non-zero digit** (e.g., in 7.00, both zeros are significant)
  - ➢ Is **explicitly marked** to indicate precision (e.g., trailing zeros in scientific notation)

Zeros are **not significant** when they:

- ✓ Are **leading zeros** before the first non-zero digit (e.g., in 0.0005010, the first four zeros are not significant—they just set the decimal point)
- ✓ Are **placeholders** for unknown or discarded digits

Examples

Number	Why These Digits Are Significant	Count
37.89	All digits are non-zero → all are significant	4
5090	5 and 9 are non-zero; the zero between them is significant; trailing zero is **not** (no decimal point)	3
7.00	7 is non-zero; both zeros are **after decimal and after non-zero** → significant	3
0.00082	Leading zeros are **not** significant; only 8 and 2 are → significant	2
0.000620	Leading zeros are **not** significant; 6, 2, and trailing 0 **after decimal** → significant	3

5.2×10^4	In scientific notation, only digits in the coefficient (5.2) count → 5 and 2	2
3.506×10	All digits in 3.506 are explicitly written → all are significant	4
8×10^{-3}	Only one digit (8) is written → only 8 is significant	1

Key Rules Recap

- ✓ **Non-zero digits** are always significant.
- ✓ **Zeros between non-zero digits** are significant.
- ✓ **Trailing zeros after a decimal point** are significant.
- ✓ **Leading zeros** (before the first non-zero digit) are **not** significant.
- ✓ In **scientific notation**, only digits in the coefficient (M in $M \times 10^n$) are counted.

1.3 Rounding Off Numbers

Rounding off is the process of reducing the number of digits in a number while keeping it as close as possible to the original value. This is often done to:

- ✓ Simplify calculations
- ✓ Match a required level of precision
- ✓ Limit the number of significant digits

Rounding-Off Rules

To round a number to **n significant digits**:

- ✓ **Drop all digits** after the nth significant digit.
- ✓ Apply these rules based on the **first discarded digit**:
  - ➢ **Less than 5** → leave the last retained digit unchanged.
  - ➢ **Greater than 5** → increase the last retained digit by 1.
  - ➢ **Exactly 5**:
    - ✓ If followed by **non-zero digits**, round **up**.
    - ✓ If followed only by **zeros**, apply the **even digit rule**:
      - ❖ If the last retained digit is **even**, leave it unchanged.

- ❖ If it is **odd**, increase it by 1.

This ensures **unbiased rounding**, especially in large datasets.

Example Set 1: Rounded to 3 and 4 Significant Figures

Original Number	Rounded to 3 Figures	Rounded to 4 Figures
00.522341	0.522	0.5223
93.2155	93.2	93.22
00.66666	0.667	0.6667

Explanation:

- **00.522341**:
  - 3 figures: Keep 5, 2, 2 → next digit is 3 → less than 5 → no change → **0.522**
  - 4 figures: Keep 5, 2, 2, 3 → next digit is 4 → less than 5 → no change → **0.5223**
- **93.2155**:
  - 3 figures: Keep 9, 3, 2 → next digit is 1 → less than 5 → **93.2**
  - 4 figures: Keep 9, 3, 2, 1 → next digit is 5 → followed by non-zero → round up → **93.22**
- **00.66666**:
  - 3 figures: Keep 6, 6, 6 → next digit is 6 → round up → **0.667**
  - 4 figures: Keep 6, 6, 6, 6 → next digit is 6 → round up → **0.6667**

Example Set 2: Rounded to 4 Significant Figures

Original Number	Rounded Value
9.6782	9.678
29.1568	29.16
8.24159	8.242 *(book typo: shows 3.142)*
30.0567	30.06

Explanation:

- **9.6782**: Keep 9, 6, 7, 8 → next digit is 2 → less than 5 → **9.678**
- **29.1568**: Keep 2, 9, 1, 5 → next digit is 6 → round up → **29.16**
- **8.24159**: Keep 8, 2, 4, 1 → next digit is 5 → followed by 9 → round up → **8.242**

The book incorrectly lists this as **3.142**, which is likely a typo.

- **30.0567**: Keep 3, 0, 0, 5 → next digit is 6 → round up → **30.06**

Why This Table Matters

It shows how:

- Rounding affects numerical precision
- Significant digits determine the accuracy of reported values
- Proper rounding helps control computational errors

1.4 Errors

In numerical analysis, an **error** is defined as:

$$\text{Error} = \text{True value} - \text{Approximate value}$$

This difference helps us understand how far off our computed or measured value is from the actual (exact) value.

1.4.1 Types of Errors

1. Inherent Errors

- ✓ These occur **in the problem statement itself**.
- ✓ Often caused by **imprecise measurements** or **uncertain data**.
- ✓ Example: If a physical constant is measured as 9.81 m/s² but the true value is slightly different, that discrepancy is inherent.

2. Analytic Errors

- ✓ Arise when converting a **mathematical model** into a **computational form**.
- ✓ Often caused by **truncating infinite series** or **simplifying assumptions**.
- ✓ Example: Approximating $\sin x$ using only a few terms of its Taylor series:

$$\sin\ x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

Discarding higher-order terms introduces analytic error.

3. Round-off Errors

- ✓ Occur due to **limited precision** in digital computations.
- ✓ Computers can't store infinite digits, so values are **rounded**.

- ✓ Example: Representing 1/3 as 0.333… truncated to 0.333 leads to round-off error.

Definitions of Error Measures

Definition 2: General Error

By the error of an approximate number we mean the difference between the exact number X, and the given approximate number x. It is denoted by E (or by Δ)

$$E = X - x$$

Where:

- X= exact value
- x= approximate value
- Eor Δ= error

Definition 3: Absolute Error

The absolute error of an approximate number x is the absolute value of the difference between the corresponding exact number X and the number x. It is denoted by EA.

$$E_A = | X - x |$$

Measures the **magnitude** of the error, regardless of direction.

Definition 4: Limiting Error

The limiting error of an approximate number denoted by Δx is any number not less than the absolute error of that number.

$$\Delta x \geq E_A$$

A **bound** on the absolute error. The true value lies within:

$$x - \Delta x \leq X \leq x + \Delta x$$

Or simply:

$$X = x \pm \Delta x$$

Definition 5: Relative Error

The relative error of an approximate number x is the ratio of the absolute error of the number to the absolute value of the corresponding exact number X, where X 0 by ER (or by δ)

$$E_R = \frac{E_A}{| X |} = \delta$$

Measures error **relative to the size of the true value**. Useful when comparing errors across different scales.

Definition 6: Limiting Relative Error

The limiting relative error δx of a given approximate number x, is any number not less than the relative error of that number

$$\delta_x \geq E_R$$

A bound on the relative error. In practice, since $X \approx x$, we often use:

$$\delta_x = \frac{\Delta x}{x}$$

Definition 7: Percentage Error

Percentage error, denoted by **Ep**, is a way to express how large the error is **relative to the true value**, but in **percentage form**. It's calculated as:

$$\text{Ep} = \text{ER} \times 100 = \frac{\text{Absolute Error}}{\text{True Value}} \times 100$$

Where:

- ✓ **ER** is the **relative error**
- ✓ **Ep** is the **percentage error**
- ✓ **Absolute Error** = $| X - x |$
- ✓ **True Value** = X

Example

If the true value $X = 22/7 \approx 3.142857$and the approximate value $x = 3.14$, then:

- ✓ Absolute Error $EA = | 3.142857 - 3.14 | = 0.002857$
- ✓ Relative Error $ER = \frac{0.002857}{3.142857} \approx 0.00091$
- ✓ Percentage Error $Ep = 0.00091 \times 100 = 0.091\%$

This means the approximation is off by about **0.09%**, which is quite accurate.

1.5 Relative Error and The Number of Correct Digits

Relative Error

Relative error measures how large the error is **compared to the true value**:

$$\text{Relative Error (ER)} = \frac{\text{Absolute Error (EA)}}{\text{True Value (X)}}$$

It helps us understand how accurate an approximation is, especially when comparing numbers of different magnitudes.

Decimal Representation of a Number

Any positive number $x$can be written as:

$$x = \alpha_m \cdot 10^m + \alpha_{m-1} \cdot 10^{m-1} + \cdots + \alpha_{m-n+1} \cdot 10^{m-n+1} + \cdots$$

Where:

- $\alpha_i \in \{0,1, \ldots ,9\}$
- $\alpha_m \neq 0$is the **first significant digit**
- This helps us locate and count significant digits

Definition 8: Correct Digits

An approximate number $x$is said to have **n correct digits** if its **absolute error** does not exceed **half a unit** in the n^{th}digit (from the left).

Example:

Let $X = 73.97, x = 74.00$

$$EA = | X - x | = 0.03 < \frac{1}{2} \cdot 10^{1-3} = 0.05$$

So, $x$is correct to **3 digits**.

Theorem: Bound on Relative Error

If a number $x$has $n$correct digits, then:

$$ER \leq \frac{1}{\alpha_m \cdot 10^{n-1}}$$

Where:

- $\alpha_m$is the **first significant digit**
- $n$is the number of correct digits

This formula gives a **maximum bound** on the relative error.

Corollary 1:

The **limiting relative error** is:

$$\delta_x = \frac{1}{\alpha_m \cdot 10^{n-1}}$$

Corollary 2:

If $n \geq 2$, then for practical purposes:

$$ER \approx \delta_x = \frac{1}{2 \cdot \alpha_m \cdot 10^{n-1}}$$

This is a simplified estimate used in real-world calculations.

1.5.1 Important Rules

These rules help estimate error based on how a number is rounded or truncated:

Rule	Condition	Error Bound		
Rule 1	Rounded to $m$decimal places	\(	X - x	\leq \frac{1}{2} \cdot 10^{-m} \)
Rule 2	Truncated to $k$digits	\(	X - x	< 10^{-k+1} \cdot X \)
Rule 3	Rounded to $k$digits	\(	X - x	< \frac{1}{2} \cdot 10^{-k+1} \cdot X \)
Rule 4	Correct to $m$significant digits	\(	X - x	< 10^{-m} \cdot X \)
Rule 5	Correct to $n$significant digits, first digit α_m	$ER < \frac{1}{\alpha_m \cdot 10^{n-1}}$		

Example 1.4:

How many digits are needed to compute 20 with ≤ 0.1% error?

- First digit of 20 is 2
- Set $ER = 0.001$
- Use Rule 5:

$$\frac{1}{\alpha_m \cdot 10^{n-1}} \leq 0.001 \Rightarrow \frac{1}{2 \cdot 10^{n-1}} \leq 0.001 \Rightarrow 10^{n-1} \geq 500 \Rightarrow n \geq 4$$

So, **4 digits** are needed.

Example 1.5:

Verify Rule 1 for $X = \frac{8}{9} = 0.888\ldots$, rounded to 3 digits: $x = 0.889$

- $EA = | X - x | = | 0.888\ldots - 0.889 | = 0.0001$

- Rule 1 says: $EA \leq \frac{1}{2} \cdot 10^{-3} = 0.0005$
- Since $0.0001 < 0.0005$, **Rule 1 is verified**

1.5.2 Tables: Error vs. Correct Digits

These tables help estimate:

- **Relative error (%)** based on number of correct digits
- **Number of correct digits** based on relative error

Table A: Relative Error (%) for n Correct Digits

First Two Digits	2 Digits	3 Digits	4 Digits
10–11	10	1	0.1
12–13	8.3	0.83	0.083
14–16	7.1	0.71	0.071
17–19	5.9	0.59	0.059
20–22	5	0.5	0.05
…	…	…	…
90–99	1.1	0.11	0.011

Table B: Correct Digits Based on Relative Error (%)

First Two Digits	2 Digits	3 Digits	4 Digits
10–11	4.2	0.42	0.042
12–13	3.6	0.36	0.036
14–16	2.9	0.29	0.029
17–19	2.5	0.25	0.025
…	…	…	…
80–99	0.5	0.05	0.005

These tables are useful for:

- ✓ Estimating how many digits are needed for a desired accuracy
- ✓ Checking if a given approximation meets error requirements

1.6 General Error Formula

In real-world computations, we often deal with a function $u = f(x_1, x_2, \ldots, x_n)$, where each input $x_i$is subject to a small error Δx_i. The goal is to estimate the resulting error Δuin the output u.

Step-by-Step Breakdown

Step 1: Define the Function

Let:

$$u = f(x_1, x_2, \ldots, x_n)$$

Each variable $x_i$has a small error Δx_i, so the perturbed value is:

$$u + \Delta u = f(x_1 + \Delta x_1, x_2 + \Delta x_2, \ldots, x_n + \Delta x_n)$$

Step 2: Apply Taylor's Theorem

Using a multivariable Taylor expansion, we approximate:

$$u + \Delta u \approx u + \frac{\partial u}{\partial x_1}\Delta x_1 + \frac{\partial u}{\partial x_2}\Delta x_2 + \cdots + \frac{\partial u}{\partial x_n}\Delta x_n$$

Neglecting higher-order terms (like $(\Delta x_i)^2$), we get:

$$\Delta u \approx \frac{\partial u}{\partial x_1}\Delta x_1 + \frac{\partial u}{\partial x_2}\Delta x_2 + \cdots + \frac{\partial u}{\partial x_n}\Delta x_n$$

This is **Formula (1)** — the **absolute error** in u.

Step 3: Relative Error Formula

Divide both sides by $u$to get the **relative error**:

$$E_R = \frac{\Delta u}{u} \approx \frac{1}{u}\left(\frac{\partial u}{\partial x_1}\Delta x_1 + \frac{\partial u}{\partial x_2}\Delta x_2 + \cdots + \frac{\partial u}{\partial x_n}\Delta x_n\right)$$

This is **Formula (2)** — the **general relative error formula**.

Interpretation

- ✓ Each term $\frac{\partial u}{\partial x_i}\Delta x_i$shows how sensitive $u$is to changes in x_i.
- ✓ The formula helps **quantify error propagation** in complex functions.
- ✓ It's especially useful in **engineering**, **physics**, and **computational modeling**, where multiple inputs affect a single output.

Summary

Concept	Formula
Absolute Error in u	$\Delta u \approx \sum_{i=1}^{n} \frac{\partial u}{\partial x_i} \Delta x_i$
Relative Error in u	$E_R = \frac{\Delta u}{u} \approx \sum_{i=1}^{n} \frac{1}{u} \cdot \frac{\partial u}{\partial x_i} \Delta x_i$

1.7 Application Of Errors To The Fundamental Operations Of Arithmetic

(i) Addition

Let:

$$u = x_1 + x_2 + \cdots + x_n$$

Each x_i is an approximate number with its own absolute error Δx_i. Then the **absolute error in the sum** u is:

$$\Delta u = \Delta x_1 + \Delta x_2 + \cdots + \Delta x_n$$

Key Insight:

The **total error in the sum** is simply the **sum of the individual errors**. This is intuitive—each uncertain value contributes to the uncertainty of the result.

Practical Guidelines for Adding Numbers with Different Accuracies

When the numbers you're adding have **different numbers of decimal places**, follow these steps to minimize error:

(a) Identify the Least Accurate Number

- Find the number with the **fewest decimal places**.
- Leave it **unchanged**—it sets the baseline for precision.

(b) Round the Others

- Round the remaining numbers to **one or two more decimal places** than the least accurate one.
- This avoids overestimating precision.

(c) Perform the Addition

- Add all numbers using the **retained decimal places**.

- Be careful to align decimal points properly.

(d) Round the Final Result

- Round the result to **one decimal less** than the least accurate input.
- This ensures the final answer reflects the true level of precision.

Example

Suppose you're adding:

- $x_1 = 345.4$(accurate to 1 decimal)
- $x_2 = 0.1834$
- $x_3 = 11.75$
- $x_4 = 0.000354$

Step-by-step:

- Least accurate: $x_1 = 345.4$→ 1 decimal
- Round others to 2 decimals:
  - $x_2 \rightarrow 0.18$
  - $x_3 \rightarrow 11.75$(already 2 decimals)
  - $x_4 \rightarrow 0.00$
- Add: $345.4 + 0.18 + 11.75 + 0.00 = 357.33$
- Final rounding: reduce to **0 decimals** → **357**

(ii) Subtraction

Let:

$$u = x_1 - x_2$$

Where:

- $x_1$and $x_2$are approximate values
- $\Delta x_1$and $\Delta x_2$are their respective absolute errors

Absolute Error in Subtraction

The error in the result $u$is:

$$\Delta u = \Delta x_1 - \Delta x_2$$

But since errors can be **positive or negative**, and we want the **maximum possible error**, we take:

$$\Delta u \approx \Delta x_1 + \Delta x_2$$

This gives a **safe upper bound** on the error in the result.

Critical Issue: Loss of Significant Digits

When $x_1 \approx x_2$, their difference $u$becomes **very small**, and many leading digits cancel out. This causes:

- **Loss of significance**: the result may have fewer correct digits than the inputs
- **Amplified relative error**: even small input errors can dominate the result

Example:

Let:

- $x_1 = 1.000$(accurate to 4 digits)
- $x_2 = 0.999$(also accurate to 4 digits)

Then:

- $u = x_1 - x_2 = 0.001$
- But the result has only **1 significant digit**, even though inputs had 4

🛠️ **Remedies to Reduce Inaccuracy**

To avoid or reduce error when subtracting nearly equal numbers:

1. Use Higher Precision

- Ensure both $x_1$and $x_2$are approximated with **sufficient accuracy**
- This helps preserve more digits in the result

2. Reformulate the Expression

- If possible, **restructure the formula** to avoid direct subtraction
- Example: Instead of computing $f(x + h) - f(x)$, use a derivative approximation or algebraic simplification

(iii) Multiplication

(a) Product of Two Numbers

Let:

$$X = x_1 \cdot x_2$$

Where:

- $x_1$and $x_2$are approximate values
- $\Delta x_1$and $\Delta x_2$are their absolute errors

Absolute Error Formula:

$$\Delta X \approx x_1 \cdot \Delta x_2 + x_2 \cdot \Delta x_1$$

This means the total error in the product is approximately the **sum of each number multiplied by the error in the other**.

Relative Error Formula:

$$ER = \frac{\Delta X}{X} \approx \frac{\Delta x_1}{x_1} + \frac{\Delta x_2}{x_2}$$

This shows that the **relative error in a product** is the **sum of the relative errors** of the factors.

(b) Product of n Numbers

Let:

$$X = x_1 \cdot x_2 \cdot \cdots \cdot x_n$$

Each $x_i$has an absolute error Δx_i

Relative Error Formula:

$$ER = \frac{\Delta X}{X} \approx \sum_{i=1}^{n} \frac{\Delta x_i}{x_i}$$

This generalizes the two-number case: the **relative error in a product of multiple numbers** is the **sum of the relative errors** of all the individual factors.

Why This Matters

- Multiplication **amplifies errors**, especially when multiple uncertain values are involved.
- This formula helps estimate how much uncertainty is introduced when computing things like:
    - Area = length × width
    - Volume = length × width × height
    - Compound interest = principal × rate × time

(iv) Division

Let:

$$X = \frac{x_1}{x_2}$$

Where:

- x_1 and x_2 are approximate values
- Δx_1 and Δx_2 are their absolute errors
- EA is the absolute error in the result X

Absolute Error Formula

The derivation is algebraically intensive, but the final simplified expression is:

$$\Delta X \approx \left| \frac{\Delta x_1}{x_2} - \frac{x_1 \cdot \Delta x_2}{x_2^2} \right|$$

This formula shows that the error in the quotient depends on:

- The error in the numerator Δx_1
- The error in the denominator Δx_2
- The values of x_1 and x_2 themselves

Relative Error Formula

The **relative error** in the quotient is approximately:

$$ER = \frac{\Delta x_1}{x_1} + \frac{\Delta x_2}{x_2}$$

This is similar to multiplication: the **relative error in division** is the **sum of the relative errors** of the numerator and denominator.

Important Notes

- Division is **more sensitive** to errors than multiplication, especially when the denominator x_2 is small or has high uncertainty.
- If x_2 is close to zero, even a small error in x_2 can cause a large error in the result.

Example Questions of Chapter 1

Example 1.6 Round-off 27.8793 correct to four significant figures.

Solution The number 27.8793 rounded-off to four significant figures is 27.88.

Example 1.7 Round-off the number 0.00243468 to four significant figures.

Solution The rounded-off number is 0.002435.

Example 1.8 Find the sum of the approximate numbers 0.348, 0.1834, 345.4, 235.2, 11.75, 0.0849, 0.0214, 0.000354 each correct to the indicated significant digits.

Solution 345.4 and 235.4 are numbers with the least accuracy whose absolute error may attain 0.1. Rounding the remaining numbers to 0.01 and adding we get

345.4 + 235.2 + 11.75 + 9.27 + 0.35 + 0.18 + 0.08 + 0.02 + 0.00 = 602.25.

Applying the even-digit rule for rounding the result we get the sum to be equal to 602.2.

∴The sum of the given numbers = 602.2.

Example 1.9 Find the number of significant figures in the approximate number 11.2461 given its absolute error as 0.25 × 10–2.

Solution Given that absolute error = 0.25 × 10–2 = 0.0025.

∴The number of significant figure is 4.

Example 1.10 Find the product 349.1 × 863.4 and state how many figures of the result are trust worthy, assuming that each number is correct to four decimals.

Example 1.10 *Find the product 349.1 × 863.4 and state how many figures of the result are trust worthy, assuming that each number is correct to four decimals.*

Solution Let $x_1 = 349.1$. $|\Delta x_1| \le 0.05$

$x_2 = 863.4$, $|\Delta x_2| \le 0.05$

and $u = x_1 x_2$

then $u = x_1 x_2 = 349.1 \times 863.4 = 301412.94$

now $$\left|\frac{\Delta u}{u}\right| \le \left|\frac{\Delta x_1}{x_1}\right| + \left|\frac{\Delta x_2}{x_2}\right| \le \left|\frac{0.05}{x_1}\right| + \left|\frac{0.05}{x_2}\right|$$

$\Rightarrow$ $$\frac{|\Delta u|}{|u|} \le (0.05)\left\{\frac{1}{|x_1|} + \frac{1}{|x_2|}\right\} = (0.05)\left[\frac{|x_1| + |x_2|}{|x_1||x_2|}\right]$$

$\Rightarrow$ $$|\Delta u| \le (0.05)\,|u|\left[\frac{|x_1| + |x_2|}{|u|}\right] = 0.05\,(|x_1| + |x_2|)$$

$\Rightarrow$ $$|\Delta u| \le (0.05)\,\lfloor 349.1 + 863.47 \rfloor = 60.6285 \simeq 60.63$$

Therefore; the true value of u lies between

$301412.94 - 60.63$ and $301412.94 + 60.63$

i.e., 301352.31 and 301473.559

i.e., 3014×10^2 and 3015×10^2

We infer that; only the first three figures are reliable.

Example 1.11 Find the difference $\sqrt{2.01} - \sqrt{2}$ to three correct digits.

Solution We know that $\sqrt{2.01} = 1.41774469 \ldots$ and $\sqrt{2} = 1.41421356 \ldots$

Let X denote the difference

$$\therefore \quad X = \sqrt{2.01} - \sqrt{2}$$
$$= (1.41774469 \ldots) - (1.41421356 \ldots)$$
$$= 0.00353$$
$$= 3.53 \times 10^{-3}.$$

Example 1.12 If $\Delta x = 0.005$ $\Delta y = 0.001$, be the absolute errors in $x = 2.11$ and $y = 4.15$, find the relative error in the computation of $x + y$.

Solution $x = 2.11,\ y = 4.15$

$\therefore$ $x + y = 2.11 + 4.15 = 6.26,$

and $\Delta x = 0.005,\ \Delta y = 0.001$

$\Rightarrow\ \Delta x + \Delta y = 0.005 + 0.001 = 0.006.$

$\therefore$ The relative error in $(x + y)$ is

$$E_R = \frac{\Delta x + \Delta y}{(x + y)} = \frac{0.006}{6.26}$$

$$= 0.000958.$$

The relative error in $(x + y) = 0.001$ (approximately).

Example 1.13

Example 1.13 Given that $u = \frac{5xy^2}{z^3}$ $\Delta x, \Delta y$ and Δz denote the errors in x, y and z respectively such that $x = y = z = 1$ and $\Delta x = \Delta y = \Delta z = 0.001$, find the relative maximum error in u.

Solution We have

$$\frac{\partial u}{\partial x} = \frac{5y^2}{z^3}, \frac{\partial u}{\partial y} = \frac{10xy}{z^3}, \frac{\partial u}{\partial z} = \frac{-15xy^2}{z^4}$$

$$\therefore\ \Delta u = \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \frac{\partial u}{\partial z}\Delta z$$

$$\Rightarrow\ (\Delta u)_{\max} = \left|\frac{\partial u}{\partial x}\Delta x\right| + \left|\frac{\partial u}{\partial y}\Delta y\right| + \left|\frac{\partial u}{\partial z}\Delta z\right|$$

$$= \left|\frac{5y^2}{z^3}\Delta x\right| + \left|\frac{10xy}{z^3}\Delta y\right| + \left|\frac{-15xy^2}{z^4}\Delta z\right| \qquad (1)$$

Substituting the given values in (1) and using the formula to find the relative maximum error we get

$$(E_R)_{\max} = \frac{(\Delta u)_{\max}}{u} = \frac{0.03}{5} = 0.006.$$

Example 1.14

If X = 2.536, find the absolute error and relative error when

I. X is rounded and

II. X is truncated to two decimal digits.

(*i*) Here $X = 2.536$

Rounded-off value of X is $x = 2.54$

The Absolute Error in X is

$E_A = |2.536 - 2.54|$

$= |-0.004| = 0.004$

$$\text{Relative Error} = E_R = \frac{0.004}{2.536} = 0.0015772$$

$$= 1.5772 \times 10^{-3}.$$

(*ii*) Truncated Value of X is $x = 2.53$

Absolute Error $E_A = |2.536 - 2.53| = |0.006| = 0.006$

$$\therefore \quad \text{Relative Error} = E_R = \frac{E_A}{X} = \frac{0.006}{2.536} = 0.0023659$$

$$= 2.3659 \times 10^{-3}.$$

Example 1.15 If π= 22 / 7 is approximated as 3.14, find the absolute error, relative error and relative percentage error.

$$\text{Absolute Error} = E_A = \left|\frac{22}{7} - 3.14\right| = \left|\frac{22 - 21.98}{7}\right|$$

$$= \left|\frac{0.02}{7}\right| = 0.002857.$$

$$\text{Relative Error } E_R = \left|\frac{0.002857}{22/7}\right| = 0.0009$$

$$\text{Relative Percentage Error } E_P = E_R \times 100 = 0.0009 \times 100$$

$$= 0.09$$

$$\therefore \quad E_P = 0.09\%.$$

Example 1.16 The number x = 37.46235 is rounded off to four significant figures. Compute the absolute error, relative error and the percentage error.

We have $X = 37.46235; \quad x = 37.46000$

$$\text{Absolute error} = |X - x| = |37.46235 - 37.46000|$$

$$E_A = 0.00235$$

$$E_r = \left|\frac{X - x}{x}\right| = \frac{0.00235}{37.46235} = 6.27 \times 10^{-5}$$

$$E_P = E_r \times 100 = 6.27 \times 10^{-3}$$